

ARUNA SENTHAMARAIKANNAN

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EXPERIENCE

Engineering Intern (AIOps)

Jan 2026 – June 2026

Driven Tech, Remote

- Architected a multi agentic SOC triage pipeline using ReAct-based Claude agents and MCP integrations, orchestrating Torq workflows across MISP, Auvik, and ConnectWise to correlate CVE exposure across 3,515 devices, reducing ticket creation from 24+ to under 6 hours
- Scaled platform to multi-enterprise by rebuilding MISP and IT-ISAC pipeline with tenant-isolated reporting, feed normalization, and automated ETL, generating 300+ weekly threat summaries with MTTA/MTTR tracking for executive visibility across 10K+ customers
- Identified manual report compilation as primary bottleneck; fine-tuned Llama 3 via LoRA on 5K+ labeled alerts with schema-constrained decoding and MLflow, automating 85% of the process and reallocating 20+ analyst-hours weekly to threat hunting; KT'd to incoming intern
- Architected LangGraph security chatbot with Azure AI Search and RAGAS evaluation pipeline that automatically scores response quality, while user thumbs up/down signals trigger periodic retraining of embedding models and reduce hallucinated outputs significantly

Graduate Research Assistant

May 2025 – Apr 2026

Northeastern University, Boston, MA

- Led 3-engineer team building (MEDAS) with doctors, architecting a super-learner integrating Gemini, Grok, GPT, and Claude with a meta-learner to dynamically weight each LLM, boosting Pass@1 from 65% to 70% and achieving 85% when at least one LLM is correct
- Built Python-based inference pipeline with FastAPI and MongoDB storing 420+ acute cases, normalizing free-text descriptions to ICD-11 using semantic retrieval and meta-learning ensemble, outperforming LLMs (58-65%) and doctors (20%) while reducing diagnostic variance
- Deployed containerized orchestrator on AWS SageMaker with parallel LLM routing and Redis caching, compressing diagnosis workflows from hours to minutes, a 60% reduction validated by clinicians; built React.js dashboard for real-time diagnosis review and validation

Assistant Civic Researcher

Jul 2025 – Dec 2025

Northeastern University, Boston, MA

- Co-developed full-stack recommendation engine for [civicXsyllabus](#) with React, Node.js, and Neo4j; designed and implemented Airtable backed CMS enabling non-technical content editors to update recommendations in real-time without engineering support; presented to 70+ attendees including state legislators at Massachusetts State House

Software Engineer

Jan 2023 – Dec 2024

Trimble Inc, Chennai, India

- Modernized legacy EJB based dispatch backend to event driven microservices with Kafka and Spring Boot for 80,000 users, implementing hexagonal architecture to isolate domain logic while using Redis for caching frequently accessed fleet data alongside existing SQL databases
- Overhauled monolithic frontend by rebuilding critical dispatcher modules in Angular 16 with lazy loading and NgRx state management, while migrating legacy React components to functional hooks and applying Atomic Design patterns for building reusable UI components
- Owned AWS EKS infrastructure using Terraform modules, set up canary deployments with automated rollbacks and LaunchDarkly feature flags, achieving zero downtime releases and eliminating production rollback failures across 200 plus enterprise tenant environments
- Diagnosed and resolved 15+ production bugs through root cause analysis and system log using DataDog, reducing recurring incidents by 40 percent, while introducing DORA metrics, building Jaeger tracing with Grafana SLO alerts, and mentoring junior developers

TECHNICAL PROJECTS

Automated Repository Documentation Pipeline

Python, FastAPI, JavaScript, GitHub Webhooks, SpringBoot

- Built an AI documentation pipeline using FastAPI and GitHub webhooks that incrementally processes modified Spring Boot services through git diff analysis, automating documentation generation across repositories and saving 6+ engineering hours per release cycle
- Implemented an LLM based validation layer that cross checks generated documentation against actual Java source code and automatically corrects inconsistent API behavior descriptions before publication, reducing documentation drift by 35% across evolving microservices

PaperRank AI

Python, JavaScript, Streamlit, Pandas, n8n, REST APIs

- Developed an AI powered research discovery platform orchestrating Ollama hosted LLMs, n8n automation workflows, and Python ranking pipelines to retrieve and rerank academic papers from natural language queries using embedding based relevance scoring
- Built a Streamlit based inference interface with Ngrok tunneling, achieving 82 percent retrieval relevance on manually evaluated research queries while enabling real time paper retrieval, reranking, and structured export workflows

CERTIFICATIONS

AWS Certified AI Practitioner

Jan 2026

Validated knowledge of AWS AI/ML services including SageMaker, Bedrock, and cloud-based model deployment workflows

TECHNICAL SKILLS

Core: Java, Python, SQL, TypeScript, Spring Boot, FastAPI, Microservices, Kafka, Redis, PostgreSQL, REST APIs
AI & LLMs: RAG, Multi-Agent Systems, LangGraph, MCP, LiteLLM, Vector Search, ChromaDB, Azure AI Search, Embeddings, RAGAS
Cloud & DevOps: AWS (EC2, S3, EKS), Docker, Kubernetes, Terraform, Kafka, Jenkins, GitHub Actions, Grafana, Prometheus
Frontend: React, Angular, Streamlit, JavaScript, Node.js
Concepts: Event-Driven Architecture, Hexagonal Architecture, System Design, CI/CD, WebSockets

EDUCATION

Northeastern University, Boston, MA

Jan 2025 – Dec 2026

Master of Science in Software Engineering Systems

GPA: 4.0/4.0

Coursework: Object Oriented Design, Advanced Data Structures, Cloud Computing, Distributed Systems, Web Design

Rajalakshmi Engineering College, Chennai, India

Aug 2019 – May 2023

Bachelor of Engineering in Computer Science and Engineering

GPA: 3.8/4.0

Coursework: Database Management Systems, Machine Learning, Computer Networks, Software Engineering